

Dyslexia Detection System Demo Deck

Text-only extraction from the PowerPoint deck.

Slide 1

Dyslexia Detection System

28-slide demonstration deck for screening, therapy, biomarkers, reports, and research planning

Built from the project's architecture, model catalog, research draft, and dashboard documentation.

Demo presentation for local use

Slide 2

Agenda

2

Problem, scope, and system overview

Data inputs, preprocessing, and model families

Screening, therapy, biomarkers, and explanation modules

Dashboards, records, evaluation, and research directions

Slide 3

Problem Statement

3

Dyslexia signals often appear across handwriting, speech, reading behavior, and eye movement rather than one isolated feature.

Single-modality screening can miss important cues and can be difficult to explain to teachers or parents.

The project is designed to support local-first, interpretable, multilingual educational assistance.

Slide 4

Why Multimodal Screening

4

Handwriting captures shape and spacing cues.

Speech captures fluency, pauses, and pronunciation quality.

Text captures reading and spelling patterns.

Behavior and gaze capture attention, hesitation, and visual search strategy.

Slide 5

Project Scope

5

Target users: students, teachers, parents, clinicians, and researchers.

Supported tasks: screening, speech therapy, visual focus testing, biomarker analysis, and report generation.

Supported languages and views: English, Bengali, and extendable multilingual workflows.

Deployment focus: local browser use, local persistence, and research-ready export.

Slide 6

System Overview

6

Input ingestion from manifest CSV, uploaded files, and live dashboard activities.

Preprocessing before branch-specific feature extraction.

Modality encoders for handwriting, audio, text, and behavior signals.

Fusion and prediction followed by explanation, intervention, and reporting.

Slide 7

Data Sources and Manifest Contract

7

Central CSV manifest keeps samples portable and reproducible.

Each sample can reference handwriting images, audio files, text samples, and behavior values.

Optional eye-tracking and biomarker tables extend the same workflow.

The manifest acts as the single source of truth for records and experiments.

Slide 8

Preprocessing Pipeline

8

Handwriting: grayscale loading, contrast normalization, and resizing.

Audio: spectral feature extraction and stable frame preparation.

Text: tokenization, padding, and language-specific vocabulary handling.

Numeric features: cleaning, scaling, and shape validation.

Slide 9

Model Catalog Overview

9

Prepared from the current repository documentation and project flow.

Slide 10

Default Multimodal Screening Model

10

The default model combines handwriting, audio, text, and behavior embeddings.

It is designed to be practical, balanced, and easy to deploy locally.

The model outputs risk or severity estimates for downstream interpretation.

Slide 11

Attention Fusion

11

Attention fusion learns which modality contributes most for a given learner.

This supports explanations such as handwriting-heavy, speech-heavy, or behavior-heavy cases.

The output is useful for both model debugging and educator-facing summaries.

Slide 12

Transformer Multimodal Variant

12

Transformer-based branches improve contextual modeling for text and longer sequences.

This variant is useful when sequence dependencies matter more than simple recurrence.

It is heavier than the baseline but can improve representation quality in research settings.

Slide 13

Vision Transformer Branch

13

Vision transformer variants split handwriting images into patches.

Patch attention helps capture layout, spacing, and structural consistency patterns.

This branch is suitable for experiments that need stronger image reasoning than CNN baselines.

Slide 14

Speech and Therapy Workflow

14

Speech therapy mode listens to a reading sample, transcribes it, and scores reading behavior.

The workflow can surface current prompt, recognized response, spelling score, and overall status.

It is intended for guided practice, not only for assessment.

Slide 15

Text and Spelling Branch

15

Character-level text modeling supports short prompts, spelling practice, and reading-response analysis.

The branch can be used alongside audio to compare spoken and written performance.

It also helps keep the system multilingual and lightweight.

Slide 16

Behavior and Eye-Tracking Signals

16

Behavior metrics include hesitation, repetition, omission, reading speed, and timing values.

Eye-tracking metrics help measure fixation, gaze control, and visual attention consistency.

These features strengthen the low-resource signal set when media inputs are limited.

Slide 17

Biomarker Discovery

17

The biomarkers page accepts structured CSV tables for analysis.

The system can detect likely label columns and highlight strong numeric signals.

This is useful for simple feature ranking and plain-language summaries.

Slide 18

Screening Output and Severity Interpretation

18

The screening result summarizes language, predicted severity, confidence, and evidence by modality.

The output is meant to read like a short educational summary, not just a raw class label.

Different views can be tuned for classroom, home, and learner-friendly messaging.

Slide 19

Explainability Outputs

19

Heatmaps and attention scores help users see why the system reached a result.

Model-level explanations can be combined with human-readable summaries.

Explainability is central to teacher and parent trust in the workflow.

Slide 20

Speech Therapy Module

20

Shows prompt, response, auto score, passing threshold, and overall status.

Supports round-based guided reading with automatic progression.

Useful for repeated practice and progress tracking across sessions.

Slide 21

Visual Focus Test

21

The visual focus test supports letters, digits, and mixed-symbol modes.

Each round asks the learner to tap a target symbol from an answer grid.

The module is designed to measure attention, speed, accuracy, and consistency.

Slide 22

Test Lab and Report

22

The test and report area collects learner details before generating the final report.

It keeps the output ready for PDF generation and session documentation.

This page is the bridge between live testing and formal reporting.

Slide 23

Records and Local History

23

Saved sessions build a local history of tests and reports.

The dashboard can show total records, filtered results, latest entry, and last saved time.

This helps the system behave like a continuing local toolkit instead of a one-off demo.

Slide 24

Evaluation Metrics

24

Prepared from the current repository documentation and project flow.

Slide 25

Deployment Surfaces

25

Streamlit dashboard

Fast research exploration and experimentation

Useful for single-user local workflows

Good for quick checks and model validation

Web dashboard

Browser-based local UI

Supports live interaction and record keeping

Best for demonstration and sharing

Slide 26

Research Opportunities

26

Collect larger multilingual datasets.

Add stronger fusion methods and self-supervised pretraining.

Validate intervention suggestions in classroom-like settings.

Extend explanation quality for teachers, parents, and learners.

Slide 27

Limitations and Ethics

27

The system is a support tool, not a diagnosis replacement.

Data quality, consent, and privacy remain essential.

Model outputs should be reviewed by trained professionals in real educational settings.

Slide 28

Conclusion

28

This project unifies screening, therapy, biomarkers, reporting, and records in one local-first platform.

The next step is to expand evaluation, strengthen data collection, and validate educational usefulness.

The deck is ready for demonstration, review, and future publication work.

Thank you